Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Lipa, John A. Ph.D.		
Project Title:	High Resolution Study of the Critical Region of Oxygen Using Magnetic Levitation		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stanford	**State:**	CA
Zip Code:	94305	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	01/31/2000	**End Date:**	01/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We have developed a large-bore solenoid system designed to apply a very uniform force to a magnetic material over a small volume. This system has been used to avoid hydrostatic compression of a fluid by cancelling the weight of each individual molecule. Critical phenomena have been studied much closer to the transition than has been possible on earth without gravity cancellation. The objective of this task is to develop an improved understanding of the phenomena near a gas-liquid critical point without the need for microgravity conditions, by using magnetic forces to cancel gravity. The material to be studied is liquid oxygen, which has an unusually high magnetic susceptibility. The results can be used to predict the expected behavior in space of other fluids which cannot be magnetically levitated. The results from this program could give us a first look at the region very close to the critical point of a gas-liquid system. These results could affect our theoretical understanding of critical phenomena as applied to a wide variety of systems. Also it could lead to guidence for future flight experiments that enter this difficult-to-study region.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 11/02/2007)